SKYLIGHT POLARIZATION FOR NAVIGATION

by

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Abstract

Co-authorship and Statement of Collaboration

Statement of Originality and Disclosure of Use of Artificial Intelligence

Acknowledgements

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List of Abbreviations

AoP angle of polarization

DoFP division-of-focal-plane

DoP degree of polarization

DoT division-of-time

GNSS global navigation satellite system

LiDAR light detection and ranging

MPSC miniaturized polarized skylight compass

RMSE root mean squared error

SNR signal-to-noise ratio

SoP state of polarization

SP4N skylight polarization for navigation

SSRM single-scattering Rayleigh model

UAV unmanned aerial vehicle

VR vortex retarder

List of Symbols

ϕ_s Solar elevation 7, 8

S Stokes vector 4–6

θ_s Solar azimuth 7, 8

Chapter 1

Introduction

1.1 Motivation

1.2 Contributions

1. Novel algorithm for extraction of navigational information from skylight polarization.

2. Experimental demonstration of polarization navigation in REALISTIC conditions.

1.3 Organization of Thesis

Chapter 2

Chapter 2

Literature Review

Almost 150 years ago, Lord Rayleigh first revealed how scattering events caused by the particles that make up the Earth's atmosphere systematically polarize the light from the sun. However, only in the past 25 years did the research community realize that Rayleigh's revelation could be leveraged for the purposes of human navigation. The laws of scattering Rayleigh describes lead to a *pattern* of skylight polarization that can be used as a navigational reference. Similar to how a magnetic compass reads the Earth's magnetic field to produce a heading, a *polarization compass* can read the Earth's skylight polarization pattern to extract navigational cues. While the research community has contributed to the realization of skylight polarization for navigation (SP4N), the natural world has been using SP4N for millennia. Researchers were first clued into this method of navigation by studying the *waggle dances* of the Honeybee. In the following chapter, we will explain the theory of atmospheric scattering that creates the skylight polarization pattern. Then, we will discuss the biological inspiration for this method. Finally, we will consider how current research is leveraging the skylight polarization pattern to create bio-mimetic navigation devices.

2.1 Representations of the State of Polarization

There are two representations of the state of polarization (SoP) that we employ in this work. The first is the intensity-angle-ratio representation. The second is the Stokes vector method. Most of the incident sun light is plane-polarized by the atmosphere which means that our representation only needs to handle plane-polarized states. That is why it is reasonable to use the intensity-angle-ratio representation even though it is a strict subset of the Stokes vector space. In the following section, we explore these two representations of the SoP of a partially plane-polarized beam.

2.1.1 Intensity-Angle-Ratio Representation

The SoP of a partially plane-polarized beam can be fully defined using an intensity, angle, and ratio. In this representation, the ratio, called the degree of polarization (DoP), is the fraction of intensity that corresponds to polarized light. The angle, called the angle of polarization (AoP), is the angle between an arbitrary reference direction and the plane of oscillation of the electric field. The AoP is also implicitly defined with respect to the direction of propagation of the beam, since we are referring to transverse polarization. Since the beam is plane-polarized, the electric field oscillates exclusively within said plane, which is the SoP only requires one angle. This representation is insufficient to encode the SoP for other modes of polarization such as elliptical. For other polarization modalities, the Stokes vector is used.

2.1.2 Stokes Vector Representation

In their 1951 paper on the polarization of the sunlit sky, Chandrasekar and Elbert acknowledge the need for a symmetrical description of the polarization state of light [1]. Up to this time, it was common to describe the polarization state of partially plane-polarized light by its intensity, degree of polarization, and angle of the plane of polarization. While this description is still useful today, encoding the polarization state with such diverse quantities as

intensity, a ratio, and an angle makes a symmetric formalization of the properties of the sky model challenging. As such, Chandrasekhar and Elbert reach for a description of the polarization state of partially plane-polarized light which lends itself well to formalization of the sky model. They reach for the Stokes vector representation. Further, this representation is useful in describing the SoP of a beam from measurements of its intensity through orthogonal linear polarizers.

A Stokes vector is a mathematical representation of the polarization state of light, first proposed by Stokes in 1851 [2]. It is denoted as S and characterized by four components S_0, S_1, S_2, and S_3 that describe the intensity and phase relationships of orthogonal polarization states. Table 2.1 contains examples of different polarization states and their representation in the Stokes formalism. It is common to visualize the Stokes vector using the geometric representation of the Poincaré sphere. Each SoP is represented by a unique point on or within the surface of a sphere where points inside the sphere correspond to partial polarization.

Plane-polarized light represents a degenerate case of the Stokes vector where $S_3 = 0$. In the study of atmospheric scattering, it is common to assume this case, since the scattering process produces a neglible circular polarization component. One benefit of this case is that the Stokes vector can be represented using the more intuitive representation in terms of intensity, DoP and AoP. Equations (2.1) and (2.2) define the AoP and DoP with respect to the Stokes vector. An alternative formulation can be made where the $1/S_0$ factor is removed from (2.2) if the original Stokes vector is normalized.

$$a = 1/2 arctan(S_2/S_1) \tag{2.1}$$

$$d = 1/S_0\sqrt{S_1^2 + S_2^2} \tag{2.2}$$

We can determine the Stokes vector of a beam by measurement of its intensity at orthogonal polarization states on the Poincaré sphere [3]. In (2.3), we determine the first three

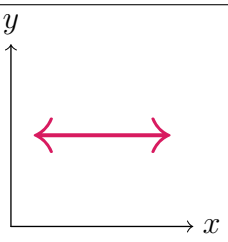
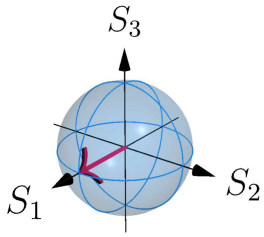
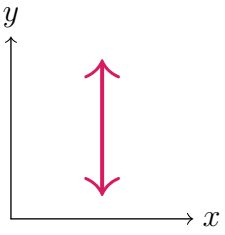
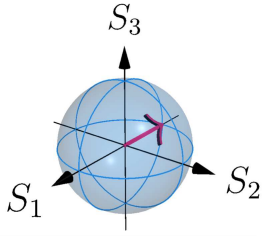
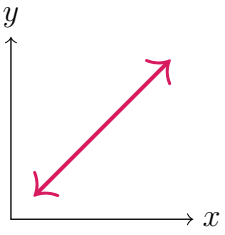
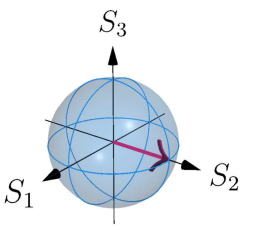
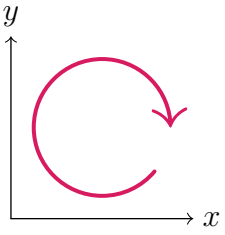
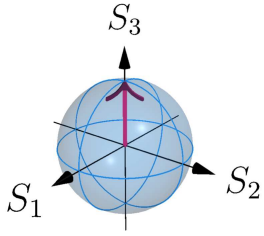
Stokes Vector	Polarization State	Poincaré Sphere
$\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$		
$\begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$		
$\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$		
$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$		

Table 2.1: Depiction of polarization states and their representation on the Poincaré sphere.

components of the Stokes vector for a plane-polarized beam. We measure the average intensity of two pairs of orthogonal, linearly polarized component waves over a measurement time much greater than the duration of one optical cycle. We denote the orientation of the component wave with the subscript angle in degrees i.e., the pairs (0°, 90°) and (45°, 135°). The circular component S_3 is undefined since we do not measure a circularly-polarized component wave. This leads to the operating principle of the division-of-focal-plane (DoFP) polarimeter whereby the intensity of four orthogonal polarization states are measured simultaneously.

$$\begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} (I_0 + I_{90} + I_{45} + I_{135})/2 \\ I_0 - I_{90} \\ I_{45} - I_{135} \\ undefined \end{bmatrix} \tag{2.3}$$

The Stokes vector provides a convenient mathematical representation for describing how physical interaction may alter the polarization state of a beam of light. The additive property of the Stokes vector means that if two independent streams of light are mixed, then the Stokes vector characterizing the mixture is the sum of the Stokes vectors of the individual streams [1]. Further, during World War II, Hans Mueller made use of the Stokes vector representation to develop his eponymous calculus [4]. Due to publication restrictions during the war, his work was only disseminated after the allied victory [5]. His insight was that the change in the SoP of a beam with Stokes vector S as a result of its linear interaction with an optical element or object via transmission, reflection, or scattering can be represented as a matrix transformation shown in (2.4) [3]. The resulting SoP after the interaction is S' and M is a 4x4 real matrix named the Mueller matrix after its inventor. The Mueller matrix represents the polarization-altering properties of the object and is a function of the frequency of light, scattering geometry, and the coordinate systems which respect to which S and S' are defined.

$$S' = MS \tag{2.4}$$

2.2 Theory of Atmospheric Scattering

In 1871, Lord Rayleigh first accounted for the principal features of the intensity and polarization of the sunlit sky with the laws of scattering that are now associated with his name [6, 7]. His work best known for its explanation of the blue color of the sky during daytime. However,

over the last century-and-a-half, it has found application in many diverse fields such as pattern recognition [8], weather analysis [9], and astronomy [10]. In recent research, Rayleigh's work has been applied to the emerging field of SP4N, which is the topic of this thesis. Such diverse applications have resulted in a variety of *sky models* which capture the effect of atmospheric scattering on impinging sunlight with different levels of complexity and fidelity. In the following section, we describe the single-scattering Rayleigh model (SSRM) using the intensity-angle-ratio representation. We justify our selection of the SSRM in consideration of alternative sky models.

2.2.1 Single-Scattering Rayleigh Model

The essential principle that underlies the Rayleigh sky model is that a scattering event polarizes the impinging beam such that the electric field vector is normal to the scattering plane. The scattering plane is defined by the observer, sun, and point where the scattering event occurs. The sun position is important because it defines the direction of propagation of the beam before the scattering event.

We show examples of the skylight polarization pattern using the SSRM in Table 2.2. The pattern is represented with one plot for the AoP and one plot for the DoP where the color is associated with a different value. Each point in a plot is mapped from a location in the sky which can be represented using an azimuth angle and an elevation angle. Different columns of the table show how the skylight polarization pattern changes for different solar elevations. The solar azimuth angle also affects the pattern, but with a rotational symmetry about the center, so variations in the solar azimuth are not shown. We show the position of the sun as a yellow circle. The solar azimuth and solar elevation angles are denoted θ_s and ϕ_s, respectively.

Fig. 2.1 depicts a scattering event that follows the Rayleigh sky model. We define an arbitrary coordinate system with basis vectors X, Y, and Z centered on the observer at O and call it the n-frame. It is common to imagine that the Earth's surface is approximately

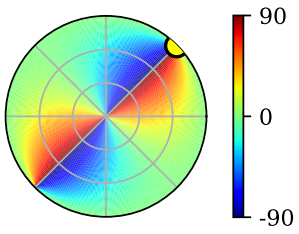
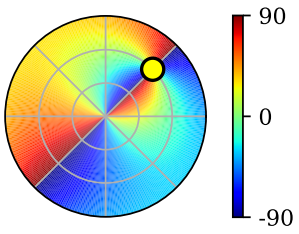
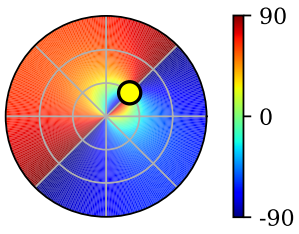
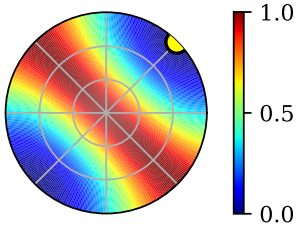
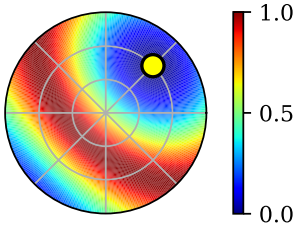
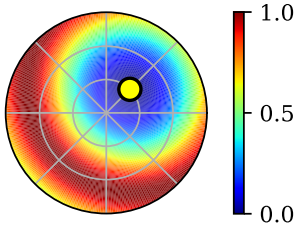
	Sun Position		
	$\theta_s = 45°\ \phi_s = 0°$	$\theta_s = 45°\ \phi_s = 30°$	$\theta_s = 45°\ \phi_s = 60°$
AoP			
DoP			

Table 2.2: Appearance of the skylight polarization pattern for different solar elevation angles.

flat for small distances around O, hence the XY plane is the ground. Then, the Z direction faces up towards the zenith of a hemispherical dome that approximates the sky as it is seen from O. We show the position of the sun S and the position of the scattering event P on the surface of this dome. These points are specified in the XYZ coordinate system using an azimuth and elevation angle. The scattering plane is depicted in Fig. 2.1 such that it contains the points O, S, and P, as previously described. The orientation of the electric field vector resulting from the scattering event at P is normal to the scattering plane. We define the electric field vector in the n-frame as $\vec{E}^n = \overrightarrow{OP} \times \overrightarrow{OS}$. Here we notice that the orientation of the electric field vector has an inherent symmetry when reflected across the scattering plane.

We can define the SoP of the beam after the scattering event using the intensity-angle-ratio representation. Consider the diagram shown in Fig. 2.2. We define the coordinate system of the beam with basis vectors X_s, Y_s, and Z_s and call it the b-frame. This coordinate system is fully defined by the selection of P with respect to the observer at O. The Z_s axis

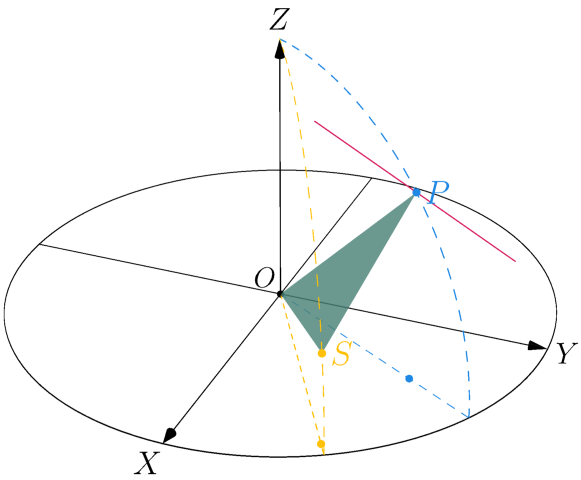


Figure 2.1: Diagram of a Rayleigh single-scattering event.

represents the direction of propagation of the beam after the scattering event. Since the beam was seen by the observer at O traveling from P, we have that $Z_s = -\overrightarrow{OP}$. The Y_s axis is perpendicular to the Z_s axis and constrained to the XY plane and the X_s axis completes the right handed coordinate system. Alternatively, the X_sY_s plane can be defined as the tangent plane to the surface of the sky dome at the point P.

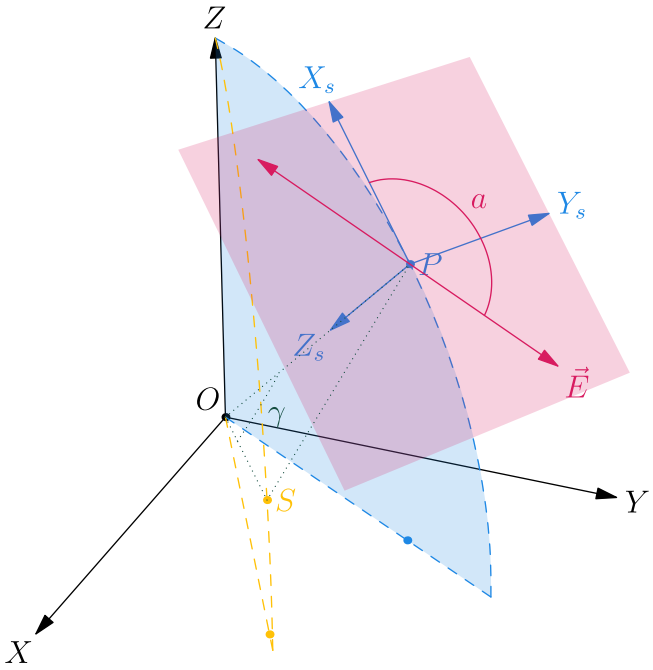


Figure 2.2: Diagram of a Rayleigh single-scattering event.

Then, the AoP, denoted a, is defined in the X_sY_s plane which is transverse to the direction

of propagation by definition. The AoP is depicted in Fig. 2.2 as a right-handed rotation from the positive X_s axis. The symmetry of the electric field vector manifests in the AoP as an ambiguity between a and $180° - a$. To resolve this ambiguity, the convention is to restrict the AoP to the range $[-90°, 90°)$.

We formalize the definition of AoP in (2.5). Given an electric field vector in the n-frame we can transform it into the b-frame as $\vec{E}^b = C_n^b \vec{E}^n$. We can convert from the n-frame into the b-frame by a transformation matrix C_n^b. The transformation matrix depends solely on the position of the point P. Then the tangent of the AoP is ratio of the x and y components of the electric field vector $\vec{E}^b$ aligned with the X_s and Y_s axes, respectively.

$$\tan a = \vec{E}_y^b / \vec{E}_x^b \tag{2.5}$$

The Rayleigh model predicts that the DoP is related to the scattering angle, denoted γ, by (2.6). The definition of the DoP according to the SSRM is simpler than the AoP since we do not have to consider the angular reference direction. Fig. 2.2 shows how we formulate the scattering angle from the position of the sun and the scattering location with respect to the observer. The scattering angle is defined in the scattering plane between the vectors $\overrightarrow{OP}$ and $\overrightarrow{OS}$.

$$d = \frac{\sin^2 \gamma}{1 + \cos^2 \gamma} \tag{2.6}$$

2.2.2 Alternative Sky Models

The SSRM makes simplifying assumptions about the scattering process. One assumption is the absence of multiple scattering in the atmosphere. Such assumptions reduce the complexity of the model and allow it to be used in low-compute applications. However, they also cause the model to diverge from empirical measurements of the skylight polarization pattern. As such, we explore other sky models which have different trade-offs associated with their predictive

capacity. We find that there are two major categories of sky model [11]. The first category is based on solving the vector radiative transfer equation through the Earth's atmosphere. In the second category, which includes the SSRM, analytic descriptions are derived from the characteristics of the skylight polarization distribution. We consider the constraints associated with both of these categories. Then, we explain why the SSRM is reasonable for SP4N applications.

Models based on solving the vector radiative transfer equation through the Earth's atmosphere can be very accurate. Examples of such models include derivations based on the matrix operator theory [12], discrete ordinate theory [13], spherical harmonics theory [14], and multiple scattering theory [15]. However, solving the vector radiative transfer equation is computationally expensive and subject to the accuracy of the initial conditions which describe, in part, the composition of the atmosphere. These factors make such models incompatible with application to navigation technologies where compute power is limited and the initial conditions are not well-defined or stable for different places on the Earth. Selection of appropriate initial conditions is a challenge for SP4N applications since different places can have different atmospheric compositions. For example, urban settings often contain aerosols generated by industrial processes which can cause a local disruption in the composition of the atmosphere. In practice, this means that the improved predictive capacity of such models is lost while the computational cost is still paid.

Models derived from the characteristics of the skylight polarization distribution provide analytic solutions which are better suited for application to navigation technologies since they are less computationally expensive than solving the vector radiative transfer equation. The SSRM is included in this category of sky model. The most prominent alternative to the Rayleigh model is the Berry model [16], but there are others [17].

According to the SSRM, there are only two neutral points in the sky occurring at the location of the sun and the anti-sun. A neutral point is a location in the skylight polarization pattern where complete depolarization occurs. The anti-sun position is directly across from

the sun position on the sky dome. Only one of these two neutral points will be visible in the sky at one time depending on whether the sun is above the horizon. In opposition to the SSRM, it has long been observed that there are actually four neutral points in the sky. By assuming that no multiple scattering happens in the atmosphere, the SSRM cannot completely describe the phenomena of *neutral points*. The lack of capacity for the SSRM to predict the location of these two additional neutral points presents a significant drawback. In 2004, Berry, Dennis, and Lee describe the Berry sky model that addresses the desire for an analytical model which captures the effect of multiple scattering, thus predicting the location of all four neutral points [16].

Despite its inability to model the effects of multiple scattering, the SSRM matches the empirical distribution of skylight polarization remarkably well [18]. Regions of the sky which are problematic for SSRM are also frequently overexposed during measurement since they fall close to the sun. In general, Suhai and Horváth find that the SSRM matches the empirical distribution better in clear sky rather than cloudy sky conditions. However, a lower solar elevation angle can help under cloudy sky conditions because the impinging skylight does not fully pass through the clouds. Instead, it can be scattered below the cloud layer in which case it is not depolarized by water clouds before reaching the measurement device. Due to its comparatively low computational cost we use the SSRM for the methods presented in this work.

2.3 Polarization-based Navigation

Polarization-based navigation is the process of extracting navigational cues from measurements of the skylight polarization pattern. Many different animal species use skylight polarization for the navigational tasks typically associated with their foraging behaviour. Taking inspiration from these species, researchers have proposed a diverse set of algorithms that enable polarization-based navigation for human purposes. We refer to such algorithms as

localization methods. However, the localization method is only a part of a complete SP4N system. We must also consider the application domain, sensor architecture, and preprocessing steps associated with a full SP4N system.

Fig. 2.3 depicts a ful SP4N system. We start with the target application domain which informs the design of the rest of the system. Within a specific domain, the sensor architecture is responsible for measurement of the skylight polarization pattern. After the polarized skylight has been measured, the preprocessing step transforms the raw measurement of polarized skylight to account for non-ideal properties of the measurement device or environment. Finally, the localization method extracts navigational cues from the measured skylight polarization pattern, which are used to form an estimate of the present state.

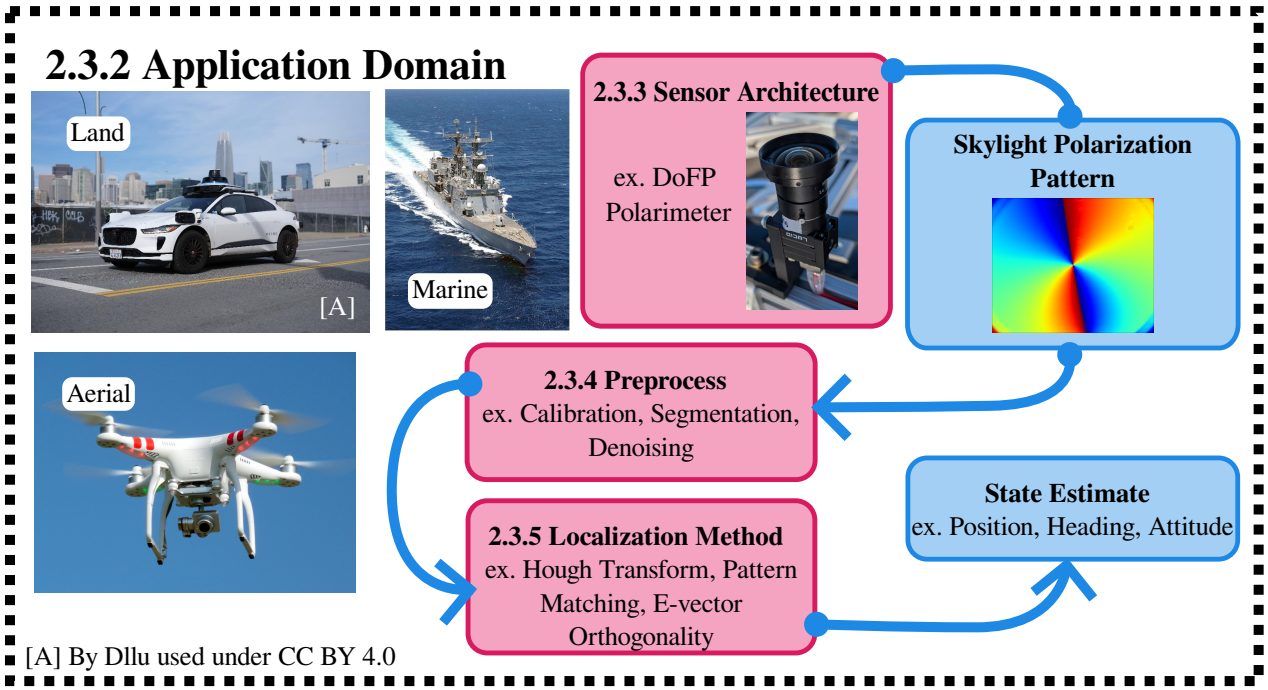


Figure 2.3: Flow diagram of a polarization-based navigation system.

This section covers each of these topics in turn. We begin with an exploration of the biological inspiration for SP4N in Subsection 2.3.1. Then, Subsection 2.3.2 presents which domains are targeted for SP4N by existing research, and includes our analysis of domain-specific challenges. In Subsection 2.3.3, we consider different sensor architectures, We consider preprocessing methods in Subsection 2.3.4 and localization methods in Subsection 2.3.5.

Finally, we compare the performance reported in recent publications in Subsection 2.3.6.

2.3.1 Biological Inspiration

The first evidence that animals employ polarization for navigation purposes comes from the study of the honeybee's *waggle dance*. In 1949, von Frisch suggested that a bee may use such dances to communicate the direction and distance of nearby food sources to other bees in its hive [19]. His seminal work tied the orientation of the waggle to the direction of a rotating linear polarizer (summarized in [20]). His experiment shows that, when a bee is made to dance on a horizontal surface in the presence of linearly polarized light from above, the orientation of the waggle changes as the direction of the polarized light is changed. The change in the orientation of the waggle suggests that honeybees can extract navigational cues from a simplified template of the skylight polarization pattern. However, it remained to be shown that foraging bees continue to use skylight polarization for navigation outside of the hive setting. More recently, experiments by Kraft et al. suggest that they indeed do [21].

Building on the early work of von Frisch, animal researchers have discovered that not only do honeybees use skylight polarization for navigation, but many other species too (summarized in [22]). North american monarch butterflies employ SP4N during their signature migration to central Mexico for overwintering [23]. During this migration, the butterfly can travel up to 4430 km [24]. At such a distance, the monarch migration is a journey for which modern aircraft require advanced navigation systems. Observed behaviours in many other species are consistent with the physiological ability to interpret the skylight polarization pattern: Desert locusts [25–27], ants [28], dung beetles [29, 30], and others. Fig. 2.4 shows example images of insects which employ skylight polarization for navigation.

In their seminal work, Lambrinos et al. gain insight about how insects employ SP4N by constructing an autonomous robot with a polarized light compass [31]. To our knowledge, this is the first example of a bionic polarized light compass in the literature. The researchers integrate a polarization compass with a small robot they call *Sahabot*. The *Sahabot* is also

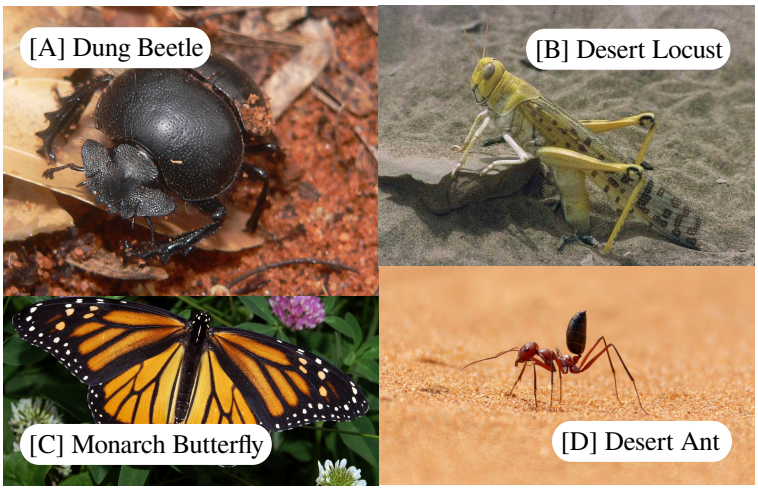


Figure 2.4: Insects that employ skylight polarization for navigation. [A] Axel Strauß, used under CC BY-SA 3.0. [C] Kenneth Dwain Harrelson, used under CC BY-SA 3.0. [D] Kingmaphotos retrieved from Encyclopædia Britannica on June 1, 2026.

given information about its state from rotatary encoders integrated with its dc motors. Lambrinos et al. test their robot on salt-pan flats in southern Tunsia under conditions which are favorable to the application of SP4N. The researchers report a best-case angular error of 0.61°. This result represents the promise of SP4N in its application to human navigation techniques.

The design of their compass follows the anatomy of the desert ant *Cataglyphis*. The eyes of *Cataglyphis* have a so-called *compound* structure made up of many eyelets called *ommatidia*. Each ommatidium is directed at a different point in space such that the total field of view can reach 180°. However, polarization vision in *Cataglyphis* is mediated by a small group of upward-facing ommatidia situated on the dorsal rim of the compound eye. The light-absorbing pigment molecules in this region cause the electrical response of the photoreceptor cells to depend on the electric field orientation of plane-polarized light. By this mechanism, *Cataglyphis* is able to read the skylight polarization pattern. The bionic measurement approach based on the compound eye structure is implemented extensively in the literature. We cover such *point* measurement methods in 2.3.3.

2.3.2 Application Domain

We find that SP4N can provide useful localization information in a variety of application domains. The performance of a specific localization methodology is dependent on its application domain. For example, localization methods which leverage the features of the local environment, such as light detection and ranging (LiDAR), will not work in a low feature environment such as marine. It is therefore important to understand the domains that researchers are considering for SP4N. We find that SP4N is tested in the land, aerial, marine, and submarine domains in order of decreasing popularity.

SP4N is used in a land setting where it is typically integrated with a four-wheeled vehicle [32–42]. The largest challenge for the land domain is obstruction of the polarization measurement by man-made or naturally occurring structures. An active area of research in this domain aims to improve the reliability of SP4N by systematic removal of such obstructions from the polarization measurement.

The aerial domain is also considered for SP4N where the most common modality is a quadrotor unmanned aerial vehicle (UAV) [43–48]. One of the largest challenges for this domain comes from the high maneuverability of aerial vehicles. Quadrotor UAVs are rarely in level flight which means that skylight polarization measurements are often skewed from the horizon. Such skewed measurements have historically been a challenge for SP4N due to their effect on the structure of the skylight polarization pattern. In many works in this domain, there is a significant correlation between the localization error and time intervals of UAV motion. Similar challenges exist in the marine domain due to wavy conditions that change the orientation of the ship deck. Recent research has focused on improving SP4N resistance to such skewed polarization measurements.

SP4N has found surprising application to the submarine domain where many other localizations methods are unreliable. The attenuation of global navigation satellite system (GNSS) signals in water makes conventional navigation solutions difficult. However, research has found that the polarization state of skylight remains useful to reasonable depths. This

opens an interesting avenue for the application of SP4N. Despite its promise, the challenges with the modulation of the polarization signal caused by interactions with the air-water boundary and non-uniformity of ocean water remain. Many recent works have developed noise reduction and filtering methods that improve the quality of the polarization measurement underwater [49–55].

2.3.3 Sensor Architecture

The literature has established many feasible sensor architectures that each have unique merits. In their 2023 review paper, Li et al. establish a taxonomy of popular sensor modalities [56]. They describe two broad classifications for polarization sensors: point and image. Point sensors use discrete photo-electronics to measure skylight polarization in a small number of view directions. Image sensors integrate a conventional camera with additional hardware to enable polarization measurement in a large number of view directions. Simply put, point sensors are extremely low resolution alternatives to image sensors. Based on the additional hardware in an image sensor, Li futher refines the classification. These subcategories describe how the polarization measurement is acheived. DoFP image sensors approach measurement by attaching a micropolarization filter array to the image sensor. Division-of-time (DoT) polarimeters capture multiple images in a short time interval, while rotating an external polarizing filter. We consider the Li framework for classifying the different sensor types.

The choice of sensor architecture has a significant impact on the downstream localization methods that are available and the performance characteristics of the full system. Point sensors are comparably low-cost and enable high polling rate. Since they only consist of discrete photo-electronics, such sensors can be constructed in a research setting easily. They are typically designed and implemented by the research team directly. The point sensor typically has fast and low-compute localization methods due to its small data volume. However, the small number of view directions in the point sensor means that it can be unreliable in

occluded conditions such as urban settings or under cloud cover. Image sensors do not suffer from the same susceptibility to occluded conditions since they capture a large number of view directions. With more view directions, there is a larger chance that clear-sky polarization measurements will still be available. This type of sensor has a large data volume which increases the complexity of the downstream localization method. Such complexity reduces the polling rate of this method over point alternatives. Despite that, image sensors are more popular due to their potential for resilient and high-precision localization.

We find that the research uses DoFP image sensors most frequently. This architecture is compact and widely available from commercial outlets. It enables the measurement of the skylight polarization pattern in a single exposure with no moving assembly which reduces the complexity of the system. Point sensors are the second most popular sensor architecture with DoT sensors used in a limited number of studies.

While the Li framework encompasses many of the fundamental sensor architectures, it does not consider the full optical system. Recent research has begun to augment the underlying sensor with additional optical elements that improve the performance of the system. The vortex retarder (VR) is a unique addition to the optical system that may support effective occlusion resistance for image sensor architectures [34, 57–59]. Ren et al. fuse DoFP and point sensors [60].

2.3.4 Preprocess

Many of the proposed localization methods rely on a *preprocess* step that transforms the raw measurement. These preprocesses are designed to improve the performance of the localization algorithm by removing artefacts from the raw measurement. We classify them into three categories: calibration, segmentation, and denoising. In some cases, researchers will combine preprocesses from more than one category.

The calibration category describes methods that measure and correct for minor defects in the measurement device typically introduced during construction. For example, image

sensors may be calibrated by measuring the camera's intrinsics which can then be used to correct for variations in the optical path of incident light [61]. Such corrections are useful since the downstream localization method can be sensitive to variations in this optical path. Other calibration procedures may involve measuring variations in the installation angle of polarizing filters [61–63]. The installation angle may deviate from the ideal design which leads to incorrect measurement of the polarization state. This can be corrected if the true installation angle is known. Vignetting is another common defect in the measurement of light intensity in a scene. The calibration of the polarization measurement device has been considered for reducing the effect of vignetting [64].

The segmentation category refers to methods that selectively remove portions of the raw measurement which do not correspond to useful skylight polarization. Segmentation is useful when portions of the measurement may be occluded by objects in the environment such as trees, buildings, and cloud cover. The challenge for segmentation procedures is choosing which portions to remove without destroying useful information [65]. Zou et al. select pixels to remove based on the orthogonality of their e-vector orientations [66]. Another popular approach uses classical edge detection algorithms like Canny [67] to remove obstructions like buildings [68]. Building surfaces like windows can reflect incident skylight while preserving its polarization state which can confuse downstream localization methods. Other segmentation approaches use DoP to remove pixels with a low signal-to-noise ratio (SNR) of the AoP.

The denoising category refers to methods which transform the measurement while preserving it for the downstream localization method. Unlike segmentation, such denoising procedures do not remove portions of the measurement. Rather, they strategically restore certain portions such that they improve the performance of the downstream localization method. There is evidence in the literature that certain animal species have evolved biological structures that perform denoising on the signals from their visual systems [69–71]. Recent work has considered how we may augment our SP4N systems with denoising algorithms inspired by this biology. Liu et al. consider the biology of the dung beetle in the design of their

denoising procedure [72]. Other works have considered more classical methods for denoising, and borrow ideas from other image processing domains [73].

2.3.5 Localization Method

The *localization method* produces a, typically partial, solution to the localization problem based on measured skylight polarization information. These methods rely on *features* of the skylight polarization pattern to extract a localization objective. The way that such features appear in a polarization measurement depends on the position, time, and orientation of the measurement device. This dependence enables SP4N.

Building a taxonomy for localization methods is challenging due to their diversity. Rather than focus on the particular algorithm, existing taxonomies classify methods based on the features of the skylight polarization pattern they employ. This approach is useful because the feature of the skylight polarization pattern is tightly coupled with the possible application scenarios. When designing an SP4N device, the application constraints can aid in determining which features should be considered which reduces the possible design space.

Kong et al. identify two popular features employed in early works [74]. The *e-vector feature* exploits the orthogonal geometry of the single scattering event. The orientation of the e-vector, propagation direction, and solar vector are orthogonal in an ideal measurement. By measuring the e-vector orientation, and knowing the propagation direction from the dynamics of the measurement device, the solar vector can be determined. The seminal work on *Sahabot* by Lambrinos et al. uses this feature [31]. The *solar meridian feature* relies on the AoP distribution along the solar meridian. The Rayleigh model predicts that these AoP values will be 90°. The angle of the solar meridian is the solar azimuth angle which can be used as a heading reference. Lu et al. use this feature in their algorithm based on the Hough transform which we implement in Section 3.2 as part of this work [75].

While Kong features are sufficient to describe many methods in the literature, it is not exhaustive. This highlights a significant gap in the existing feature-based taxonomy for lo-

calization methods. Futher, the features defined in the taxonomy are not completely distinct which can make some methods ambiguous. Zhao et al. identify a property of the e-vector orientation they call zenith circle additivity [48]. In brief, the mean of the e-vector orientations for any circle of rays centered about the zenith is equal to the solar azimuth. We classify their method with the e-vector feature since this property is an extension of the orthogonality of the e-vector. However, it is not clear whether the zenith circle additivity should be a distinct feature since other features could be considered extensions of the e-vector feature themselves. We propose an extension to the Kong taxonomy that includes additional features discovered in the literature.

The DoP measures the uniformity of the e-vector orientation in a ray of light. It is most frequently employed in a preprocess procedure to remove portions of a measurement that do not provide useful polarization information. In this application, DoP is not considered a feature, since it does not provide navigational cues. However, the distribution of DoP is well-defined for a given position, time, and orientation which makes it possible to use as a feature. Kronland-Martinet et al. use this feature to determine the solar vector and subsequently the north celestial pole [76]. In this case, we consider it a feature since they show that its distribution can be used to find the position of the sun in the sky, which is a partial solution to the localization problem.

The application of DoP as a feature is relatively unpopular, with researchers preferring AoP features. This is due, in large part, to DoP being widely considered less stable than the AoP distribution under occlusions. This is because light interactions with many potential obstructions frequently has a depolarizing effect that changes the DoP. AoP does not have the same problem because relatively fewer potential obstructions can alter it.

In the sky, there are four loci called *neutral points* which undergo significant depolarization [16]. Such neutral points can be determined using the AoP and DoP distributions [77]. These points can provide a strong navigational cue which has been used in various localization methods [68, 78–80].

We also find that methods may employ more than one feature to determine a solution to the localization problem. As a representative case, we consider the *model optimization* procedure [38]. The pattern matching algorithm that we propose in this thesis is an example of model optimization. Such methods represent the localization problem as an optimization problem where estimated states are evaluated against a sky model. By minimizing the difference between the measurement and the sky model, an accurate estimate of the state can be achieved. Since these approaches employ a full sky model, they do not consider only a single feature.

2.3.6 Comparison of Methods

We compare the results from recent publications that develop polarization-based navigation solutions for heading estimation in Table 2.3. We classify the full SP4N system by the application domain, sensor architecture, preprocessing steps, and whether or not the polarization sensor was fused with other sensors. We report the heading root mean squared error (RMSE) for each publication taking the best case where there are multiple similar experiments. In publications where the polarization sensor is fused, we try to report the error from the polarization sensor if it is available.

Despite controlling for different classifications of polarization system, the heading error is challenging to compare directly. This is in large part due to the lack of a widely adopted benchmark dataset in the SP4N field. The dataset provides important context for interpretation and comparison of a method's performance. However, most publications develop unique datasets that vary in difficulty which affects the reported error. As such, we select the best examples from Table 2.3 and provide context for their result based on their experimental procedure. Based on this analysis, we also highlight opportunites for further development and make a case for the methods proposed in this work.

Ensure domain, sensor, and error are listed in the table or explain why they aren't.

Eschelman and Smith present results from their complete SP4N system called SkyPASS

developed in an industrial context by Polaris[1] [81]. They claim that SkyPASS can achieve a 0.1125° RMSE for heading determination in clear to partly cloudy skies. While this represents a highly accurate polarization-based navigation sensor, they provide minimal explanation of their localization method, and the environment in which their results are collected. It is not clear whether the error they present is from a dynamic test, such as when mounted on a vehicle, or a fixed test where the device is mounted to a static rotating platform. Dynamic tests are considerably more challenging for polarization-based navigation solutions. Further, the error they present is from the SkyPASS system as a whole, which includes fused inertial sensors, rather than exclusively via estimations from polarized skylight. The lack of detail in their methodology and testing procedure makes SkyPASS difficult to compare against other approaches.

Huang et al. develop a miniaturized polarized skylight compass (MPSC) for heading determination using a custom sensor architecture based on a VR [34]. They report a heading error of 0.14° RMSE. The dataset that they use for their dynamic testing approximates an ideal environment for polarization-based navigation. The sun has a low elevation angle of 16° due to the experiment taking place in the evening between 17:30 and 18:00, which improves the DoP in the zenith region of the sky. The weather during the experiment is sunny and cloudless. The vehicle moves at slow and constant speed of $5.4\,\mathrm{km\,h^{-1}}$. The trajectory is only 40 m long over a smooth road surface with gentle, regularly spaced turns. Their work does not consider the challenges associated with a realistic dataset which is a significant research objective in this thesis.

[1] `https://www.polarissensor.com`

Table 2.3: Comparison of polarization-based navigation solutions for heading estimation.

Reference	**Domain**	**Sensor**	**Preprocess**	**Fused?**	**Heading RMSE**
[81]	F, A	DoFP	S	✓	0.11
[34]	L	VR	S		0.14
[42]	L	Point		✓	0.15
[45]	A	Point		✓	0.24
[48]	A	DoFP			0.26
[43]	A	DoFP		✓	0.27
[39]	L			✓	0.28
[37]	L	DoFP	S, D		0.34
[82]		DoFP	D		0.41
[38]	L	DoFP	S		0.43
[47]	A	DoFP			0.55
[66]		DoFP			0.56
[83]	L, SM	DoFP			0.56
[72]		DoFP	S, D		0.64
[68]	M	DoFP	S, D	✓	0.64
[84]	F	DoFP	S		0.70
[44]	A	DoFP		✓	0.86
[85]		DoT	S		1.01
[33]	L	DoFP	S		1.01
[57]					1.01
[46]	A	DoFP	S, D	✓	1.14
[35]	L	DoFP		✓	2.40
[40]	L	DoFP			2.75
[36]	L	DoFP		✓	
[41]	L	DoFP	D, C		

C: Calibration, S: Segmentation, D: Denoising
F: Fixed, L: Land, A: Aerial, M: Marine, SM: Submarine

Chapter 3

Methodology

3.1 Instrumentation

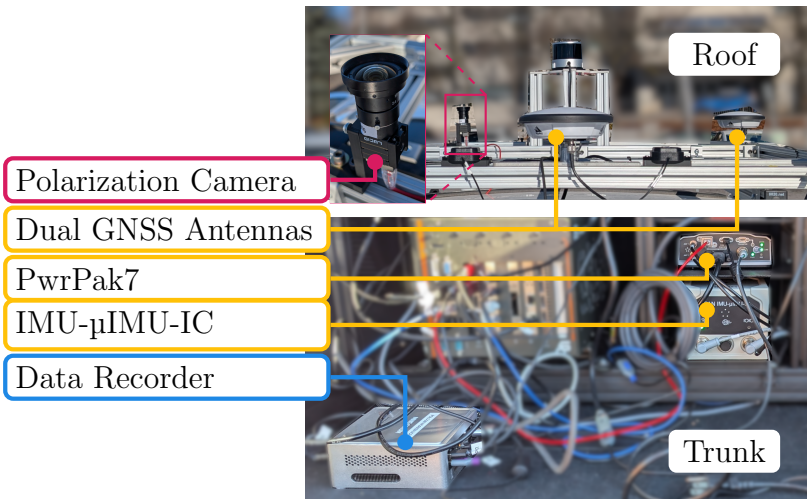


Figure 3.1: Diagram of the experimental instrumentation we use to capture the skylight polarization dataset. The polarization camera is highlighted with a close-up image for clarity. Instruments that are part of the reference system are distinguished using the yellow color.

3.2 The Hough Transform

3.3 The Pattern Matching Algorithm

Chapter 4

Experimental Demonstration

Chapter 5

Conclusion

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